

Lab 5: Data from Multiple Tables (8% of total grade)

Submission: Use the included .sql file to put your answers in, then upload only this PDF file to Blackboard (Assessments > Lab 5 - Multiple Tables).

Name your file: HTTP5126-L5-MultipleTables-LastNameFirstName.pdf, replace LastNameFirstName with your name as displayed in Blackboard.

Purpose: To practice data retrieval from multiple tables by joining tables in result sets.

Requirements: For this assignment, you will use the **updated** Pet Store data tables.

NOTE: Run your queries on your database to make sure desired results are retrieved. Also import and execute your sql file to ensure it runs all your queries before submitting.

Pre-Lab:

1. Start your mySQL server and open phpMyAdmin.
2. Create a database for this lab (eg. http5126_lab5). Set the collation as utf8_unicode_ci.
3. Select the DB then import and execute the SQL file 'lab5_pet_store_tables.sql'. This should create 3 tables in your database (employee, stock_item, & sale) which will be used for the queries you create below.

Part 1: What are the sales for a particular item? (1%)

- A. Select all of the rows from the sale table with an stock item id value of 1014.
- ```
SELECT Sale.*, Stock.name, Stock.price, Stock.category FROM sales AS Sale INNER JOIN stock_items AS Stock ON Stock.id = Sale.stock_item_id WHERE Sale.stock_item_id = 1014;
```
- B. Select date (from sale), item name (from stock\_item) to get the items sold with an id value of 1014.
- ```
SELECT Item.id AS 'Item ID', Sale.date, Item.name, Item.price FROM sales AS Sale INNER JOIN stock_items AS Item ON Item.id = Sale.stock_item_id WHERE Sale.stock_item_id = 1014;
```

Part 2: What are the sales for a particular team member? (1%)

- A. Select all of the rows from the sale table with an employee id value of 111.
- ```
SELECT s.*, e.first_name, e.last_name, e.role FROM sales AS s INNER JOIN employees AS e ON e.id = s.employee_id WHERE e.id = 111 ;
```
- B. Select date (from sale), first and last name (from employee), item id (from sale) to show the sales for the employee with an id of 111.
- ```
SELECT s.date, e.first_name, e.last_name, s.stock_item_id FROM sales AS s INNER JOIN employees AS e ON e.id = s.employee_id WHERE e.id = 111;
```

Part 3: Recent sales (1%)

- A. List the date (from sale) and item name (from stock_item) for the last 5 items sold. Ensure your results show the most recent sale first.

```
SELECT s.date, i.name, i.price FROM sales as s INNER JOIN
stock_items AS i ON i.id = s.stock_item_id ORDER BY s.date DESC
Limit 5;
```

- B. For the 5 most recent sales, display the first and last name of the employee (from employee) who processed them. Also include the date of the sale (from sale) ordered from most recent to least recent. And the name and price of the item (from stock_item).

```
SELECT s.date, e.first_name, e.last_name , i.name, i.price
FROM sales as s INNER JOIN employees AS e ON e.id =
s.employee_id INNER JOIN stock_items AS i ON i.id =
s.stock_item_id ORDER BY s.date DESC Limit 5;
```

Part 4: What a week! and Go Team! (2%)

- A. Manager: "I need to review sales activity for a specific period. Can you provide me with a list of all sales made between January 12 and January 18 (inclusive) of this year? The report should include the date of each sale, the item sold, and the first name of the employee who processed the transaction."

```
SELECT s.date, i.name, e.first_name FROM sales AS s INNER JOIN
stock_items AS i ON i.id = s.stock_item_id INNER JOIN employees
AS e ON e.id = s.employee_id WHERE s.date >= '2025-01-12' AND
s.date < '2025-01-19';
```

- B. Manager: "I want to see how well each salesperson is performing. Can you provide a report that shows the total number of sales made by each employee, grouped by their first and last name? The results should be ordered from the highest number of sales to the lowest."

```
SELECT CONCAT(e.first_name, " ", e.last_name) AS Name, count(e.id) AS
total_sales FROM employees as e INNER JOIN sales AS s ON s.employee_id
= e.id GROUP BY Name ORDER BY total_sales DESC;
```

Part 5: Challenge (2%)

- A. Manager: "I want to take a closer look at our top-performing salesperson. For all sales made by the top-seller, provide the date of the sale, the item name, its price, and category, along with the employee's first name." *Use aliases to rename your tables so your query is shorter to write. For this and all future labs you may use these table aliases in queries if you would like to*

```
SELECT
```

```

s.date,
i.name AS item_name,
i.price,
i.category,
e.first_name
FROM sales AS s
INNER JOIN employees AS e ON e.id = s.employee_id
INNER JOIN stock_items AS i ON i.id = s.stock_item_id
WHERE s.employee_id = (
    SELECT employee_id
    FROM sales
    GROUP BY employee_id
    ORDER BY COUNT(*) DESC
    LIMIT 1
)
ORDER BY s.date;

```

- B. Manager: "I need a report that shows the 'Times Sold' for each item. Please include the item ID, item name, price, and category for each item, even if it hasn't been sold. The results should be ordered by item ID."

```

SELECT i.id,i.name,i.price,i.category ,COUNT(*) AS Total_sales FROM
sales AS s LEFT JOIN stock_items as i ON s.stock_item_id = i.id
GROUP BY i.id ORDER BY i.id DESC;

```

Part 6: Challenge Yourself (1%)

Create your own "human focused question" with the data set provided. Then write a query to answer the question. The question should be written as if you are either a customer of the store or an employee of the store (Manager, Sales Associate, Stock Person). Refer to the way some of the questions are written in this and earlier labs for an idea of how to write your "human focused question".

Your query should use at least 1 **JOIN** and should be **ordered** so data is easy to parse. Preferably you should use an **aggregate function** to make your query more interesting.

- A. Write the question you have created next to the comment on line 35 of the provided sql file. eg. "—Give me the employee list, id, first name, role and their total sales
B. Create the query that returns the result your question asks for.

```

SELECT e.id,e.first_name,e.phone,e.`role` ,COUNT(e.id) AS
total_sales FROM employees AS e LEFT JOIN sales AS s ON
s.employee_id = e.id GROUP BY e.id ORDER BY total_sales DESC ;

```